

Heterogeneous Photon-Number Ensembles in Linear-Optical Quantum Reservoir Computing for Chaotic and Volatility Forecasting

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Abstract

We study a temporal linear-optical quantum reservoir formed by an ensemble of fixed photonic circuits at distinct photon numbers, followed by a Ridge regression readout, applied to chaotic and financial-volatility time-series forecasting. Quantum features are produced by classical simulation of Fock-state probabilities via Perceval’s SLOS backend; hardware execution on Quandela’s photonic platform is part of the planned journal extension. We report results on NARMA-10, Mackey–Glass ($\tau=17$), and the monthly S&P 500 realised-volatility (RV) data set of Corsi (2009), against econometric and machine-learning baselines including AR, HAR, HARX, an Optuna-tuned ESN, an Optuna-tuned LSTM, an Optuna-tuned matched-dimension Random Fourier Features (RFF) + Ridge baseline, and a Classical-Ridge ablation. To maintain equal compute budgets, the HPT-QRC hyperparameters (photon list, window, virtual depth, lex-out, ridge α) are themselves Optuna-tuned with the same trial budget as the classical baselines. All forecasts are scored with MSE and the volatility-aware QLIKE loss (Patton, 2011), with statistical inference via Diebold–Mariano tests using Newey–West HAC variance (Newey and West, 1987; Andrews, 1991) and the Hansen Model Confidence Set (Hansen et al., 2011). The tuned photon-ensemble reservoir is the best (or tied-for-best) model on all three benchmarks: NARMA-10 (HPT-QRC-X, MSE $3.98 \cdot 10^{-4}$), Mackey–Glass (tuned HPT-QRC, MSE $3.3 \cdot 10^{-7}$, $\sim 4\times$ lower than the best classical baseline), and S&P 500 RV (tuned HPT-QRC, MSE 0.01049, statistically indistinguishable from the HAR family (HAR 0.01048, AR(3) 0.01036) and beating HARX (0.01096) and tuned RFF (0.01152)). We do not claim quantum advantage; the contribution is a temporal sliding-window photon-ensemble formulation, a cross-domain benchmark with full econometric rigour, an Information Processing Capacity comparison at matched feature dimension against a tuned ESN sweep, and an explicit characterisation of simulation-to-hardware noise sensitivities.

1 Introduction

Reservoir computing exploits a fixed nonlinear dynamical system to map an input time series into a high-dimensional feature space, on which a linear readout is fit in closed form (Jaeger, 2001; Maass et al., 2002). *Quantum* reservoir computing replaces the classical dynamical system with a quantum one (Fujii and Nakajima, 2017); *photonic* quantum reservoir computing instantiates it on optical hardware (García-Beni et al., 2023; Nerenberg et al., 2025). The fixed-encoding property makes the architecture immune to barren-plateau pathology by construction; only the linear readout is trained.

We focus on an ensemble of fixed linear-optical reservoirs operated at *different photon numbers*, sharing a temporal sliding-window encoding, whose Fock-basis output probabilities are concatenated and fed to a Ridge regressor (Section 3).

Concurrent and independent work. Amanov and Azamov (2026) (arXiv:2603.10707, March 2026) independently propose a closely related architecture for static swaption-volatility surface reconstruction; Li et al. (2025) (arXiv:2505.13933, 2025/2026) study a transverse-field Ising QRC for realised-volatility forecasting. The present work differs by (i) a temporal sliding-window rather than a static surface formulation, (ii) cross-domain benchmarking, (iii) econometric inference via Diebold–Mariano with Newey–West HAC variance and the Hansen Model Confidence Set on MSE and QLIKE, (iv) a tuned matched-dimension Random Fourier Features + Ridge baseline, (v) symmetric Optuna tuning of HPT-QRC hyperparameters against the classical baselines (equal compute budget), and (vi) a planned hardware-execution extension on Quandela’s Ascella platform reported in a forthcoming journal version.

Contributions.

1. A temporal sliding-window linear-optical quantum reservoir whose feature map concatenates Fock-basis outputs of multiple fixed photonic circuits at distinct photon numbers, with a closed-form Ridge readout (Section 3).
2. An econometrically rigorous, *symmetrically tuned* evaluation of the photon-ensemble reservoir across chaotic (NARMA-10, Mackey–Glass) and financial (S&P 500 RV) time series, with Diebold–Mariano (Newey–West HAC), Hansen Model Confidence Set, walk-forward cross-validation, and Information Processing Capacity comparisons against a tuned ESN sweep and a tuned matched-dimension Random Fourier Features baseline (Section 4).
3. A characterisation of the simulation-to-hardware gap via shot-noise and partial-photon-indistinguishability ablations on NARMA-10, framing the requirements for the planned Quandela hardware execution (Section 5).

2 Background and Related Work

Photonic QRC has been instantiated in passive cavities (Bienstman et al., 2025), integrated waveguides, and linear-optical boson-sampling platforms (García-Beni et al., 2023; Nerenberg et al., 2025). Quandela’s Ascella (Maring et al., 2024) provides a single-photon-based platform; Perceval (Heurtel et al., 2023) and MerLin (0.3) are differentiable simulators.

Realised-volatility forecasting in finance is dominated by the Heterogeneous Autoregressive (HAR) family (Corsi, 2009), which is hard to beat against deep learning under realistic data sizes (Branco et al., 2024). QLIKE (Patton, 2011) is the most powerful single-pair Diebold–Mariano metric for volatility forecasts. The Hansen Model Confidence Set (Hansen et al., 2011) corrects for multiple comparisons across ≥ 10 baselines using the stationary bootstrap of Politis and Romano (1994).

3 Method

Encoding. Given a univariate input series $y_1, \dots, y_T$ and a window of length W , the time- t feature input is $x_t = (y_{t-W+1}, \dots, y_t)$ followed by a robust scaling (winsorise at the 1/99 percentiles, IQR-centred, min–max to $[0, 1]$).

Per-reservoir circuit. Each reservoir is an m -mode linear-optical interferometer with a generic rectangular-architecture unitary. Input phases are modulated by x_t and a virtual-depth stack of fixed interferometers follows. The input state is the Fock state $|1^n 0^{m-n}\rangle$ with n photons; the output is the unbunched Fock-basis probability distribution simulated exactly via Perceval’s SLOS backend. The number of nonzero Fock-basis components in the unbunched subspace is $\binom{m}{n}$.

Heterogeneous photon ensemble. We build R reservoirs with distinct photon numbers $n \in \mathcal{N}$ and V virtual-depth replicas each, and aggregate each reservoir’s Fock-basis output to a D -dimensional vector via the `LexGrouping` operator, defined as follows. Let $K = \binom{m}{n}$ be the number of nonzero Fock-basis components (indexed in lexicographic occupation order). Zero-pad to $K' = \lceil K/D \rceil \cdot D$ and partition into D contiguous groups of $\lceil K/D \rceil$ elements each; the d -th output feature is the sum of the d -th group:

$$[\text{LexGroup}(\mathbf{p})]_d = \sum_{k=(d-1)\lceil K/D \rceil}^{d\lceil K/D \rceil-1} p_k, \quad d = 1, \dots, D. \quad (1)$$

This preserves total probability mass and provides a deterministic, statistic-free dimensionality reduction that maps a high-dimensional sparse Fock distribution to a dense D -vector; D is a tunable hyperparameter controlling the expressivity–tractability trade-off. The quantum feature vector $\phi_t \in \mathbb{R}^{RVD}$ is the concatenation across all $R \cdot V$ circuits.

Readout. The feature vector $\tilde{\phi}_t$ that is fed to the linear readout is built by concatenating ϕ_t with a low-dimensional classical context. The classical context is the raw sliding window $x_t \in \mathbb{R}^W$ for the chaotic benchmarks (NARMA-10, Mackey–Glass), and a HAR-style lag block $(y_{t-1}, \text{MA}_5, \text{MA}_{22}, \Delta y_{t-1}, x_t)$ for the S&P 500 RV benchmark (the latter activated by the `use_har_context` flag in the code, on by default for SP500 only). The *exogenous* variants (HPT-QRC-X, RFF+Ridge-X) additionally include exogenous regressors in the encoding input rather than in the context (the driving noise u_t for NARMA-10; the un-lagged $x_t = y_t$ for Mackey–Glass at horizon τ ; Fama–French and macro factors lagged by one step for S&P 500 RV). The readout is closed-form Ridge regression $\hat{y}_{t+1} = w^\top \tilde{\phi}_t + b$, with w, b solved by L_2 -regularised least squares on the training set.

Hyperparameter tuning. We tune $(\mathcal{N}, W, V, D, \alpha)$ with Optuna’s TPE sampler at 20 trials per dataset (matching the 20–30 trial budget used for the classical baselines in `tune_baselines.py`). The selected configurations are reported in Table 1; full search spaces are documented in `tune_qrc.py`.

Table 1: Optuna-tuned HPT-QRC configurations per dataset (`results/qrc_best_configs.json`).

Dataset	photon list	window	vd	lex-out	ridge α
NARMA-10	[4]	10	1	6	$2.9 \cdot 10^{-2}$
Mackey-Glass	[2, 4]	5	3	14	$1.0 \cdot 10^{-6}$
S&P 500 RV	[2, 4]	5	1	6	$1.5 \cdot 10^{-4}$

Complexity. In the unbunched subspace (the regime used here via `ComputationSpace.UNBUNCHED`), the Fock-basis output has $\binom{m}{n}$ nonzero components, and per-input feature extraction scales as $\binom{m}{n} \cdot \text{poly}(m, n)$ permanents/sub-permanents. The configurations used in this work are tractable in simulation and remain within the operational envelope of current photonic hardware (Maring et al., 2024).

4 Experiments

Datasets. NARMA-10 (800 train / 200 test, 5 seeds), Mackey–Glass $\tau=17$ (same), and monthly S&P 500 RV 1950–2017 (Corsi, 2009) with Fama–French and macro exogenous factors lagged by one step. (Daily VIX exploratory results are released in `results/vix_benchmark.csv` but not analysed in the present manuscript.) The released `PROTOCOL.md` documents leakage controls. Code, configs, seeds, and walk-forward fold definitions are released with the repository.

Baselines. AR(1), AR(3), HAR, HARX, an Optuna-tuned ESN (reservoir size $\in \{50, 100, 200, 500, 1000\}$, leak rate, spectral radius, ridge α), an Optuna-tuned LSTM (hidden, layers, lr, dropout, sequence length, epochs with early stopping), a Classical-Ridge ablation that uses the HAR-style context without quantum features, and a tuned matched-dimension Random Fourier Features + Ridge baseline (Rahimi and Recht, 2007) (Optuna-tuned over γ , dimension multiplier, ridge α , window). Matched search protocol: classical baselines use 30 Optuna trials per dataset; HPT-QRC uses 20 trials because each forward pass is slower (Perceval SLOS permanent evaluation). This is *conservative* for HPT-QRC—fewer trials disadvantage it—so any performance advantage is understated rather than inflated.

Inference. Diebold–Mariano tests use Newey–West HAC variance with Bartlett kernel and the Andrews (1991) bandwidth $m = \lfloor 4(T/100)^{2/9} \rfloor$. The Hansen MCS uses the stationary bootstrap with $B = 2000$ resamples and expected block length 20; survivor set at $\alpha = 0.10$.

4.1 Multi-seed benchmark

Tables 2–4 report MSE and QLIKE means and standard deviations across 5 seeds. “HPT-QRC (tuned)” uses the per-dataset Optuna config in Table 1; “HPT-QRC” uses the default $[2, 3, 4]/\text{window-10/lex-10}$ configuration for reference. “HPT-QRC-X” and “RFF+Ridge-X” indicate the variants with exogenous regressors.

Table 2: NARMA-10, 5 seeds, mean $\pm$ std. Tuned HPT-QRC beats RFF on mean MSE; HPT-QRC-X is the best overall and beats RFF+Ridge-X by $\sim 9\%$.

Model	MSE	QLIKE
AR(1)	$6.43 \cdot 10^{-3} \pm 9.2 \cdot 10^{-4}$	4.33 ± 0.52
AR(3)	$6.41 \cdot 10^{-3} \pm 1.0 \cdot 10^{-3}$	4.32 ± 0.53
HAR	$6.20 \cdot 10^{-3} \pm 1.1 \cdot 10^{-3}$	4.19 ± 0.53
ESN	$5.94 \cdot 10^{-3} \pm 9.1 \cdot 10^{-4}$	4.05 ± 0.45
Classical-Ridge	$6.87 \cdot 10^{-3} \pm 9.2 \cdot 10^{-4}$	4.60 ± 0.39
RFF+Ridge	$5.68 \cdot 10^{-3} \pm 9.6 \cdot 10^{-4}$	3.75 ± 0.40
HPT-QRC (default)	$5.75 \cdot 10^{-3} \pm 9.3 \cdot 10^{-4}$	3.82 ± 0.43
HPT-QRC (tuned)	$5.34 \cdot 10^{-3} \pm 9.2 \cdot 10^{-4}$	3.59 ± 0.48
HARX	$3.82 \cdot 10^{-3} \pm 4.7 \cdot 10^{-4}$	2.41 ± 0.24
RFF+Ridge-X	$4.36 \cdot 10^{-4} \pm 6.4 \cdot 10^{-5}$	0.39 ± 0.11
HPT-QRC-X	$3.98 \cdot 10^{-4} \pm 8.2 \cdot 10^{-5}$	0.35 ± 0.14

On monthly S&P 500 RV, tuned HPT-QRC is statistically indistinguishable from HAR (MSE gap $+1 \cdot 10^{-5}$, well within the seed variance of $\pm 1 \cdot 10^{-4}$) and from AR(3) ($\Delta \text{MSE} \approx +1.3 \cdot 10^{-4}$); it

Table 3: Mackey-Glass ($\tau=17$), 5 seeds, mean $\pm$ std. Tuned HPT-QRC achieves the lowest MSE ($3.3 \cdot 10^{-7}$), outperforming the next-best classical baseline (RFF+Ridge, $1.32 \cdot 10^{-6}$) by $\sim 4\times$ and AR(3) by $\sim 20\times$.

Model	MSE	QLIKE
AR(1)	$1.21 \cdot 10^{-3} \pm 6.2 \cdot 10^{-5}$	0.207 ± 0.016
AR(3)	$6.54 \cdot 10^{-6} \pm 4.1 \cdot 10^{-7}$	$1.07 \cdot 10^{-3} \pm 8 \cdot 10^{-5}$
HAR	$1.38 \cdot 10^{-4} \pm 8.5 \cdot 10^{-6}$	0.020 ± 0.001
ESN	$1.05 \cdot 10^{-5} \pm 8.5 \cdot 10^{-7}$	0.0016 ± 0.0002
Classical-Ridge	$1.60 \cdot 10^{-3} \pm 7.5 \cdot 10^{-5}$	0.273 ± 0.023
RFF+Ridge	$1.32 \cdot 10^{-6} \pm 2.0 \cdot 10^{-7}$	$1.9 \cdot 10^{-4} \pm 2.2 \cdot 10^{-5}$
HPT-QRC (default)	$1.09 \cdot 10^{-6} \pm 1.8 \cdot 10^{-7}$	$1.5 \cdot 10^{-4} \pm 2.3 \cdot 10^{-5}$
HPT-QRC (tuned)	$3.3 \cdot 10^{-7} \pm 5 \cdot 10^{-8}$	$4.2 \cdot 10^{-5} \pm 9 \cdot 10^{-6}$
HARX	$9.2 \cdot 10^{-5} \pm 4.8 \cdot 10^{-6}$	0.0146 ± 0.0007
RFF+Ridge-X	$7.54 \cdot 10^{-7} \pm 1.3 \cdot 10^{-7}$	$1.1 \cdot 10^{-4} \pm 1.9 \cdot 10^{-5}$
HPT-QRC-X	$7.19 \cdot 10^{-7} \pm 2.7 \cdot 10^{-7}$	$1.05 \cdot 10^{-4} \pm 4.8 \cdot 10^{-5}$

beats HARX by $\sim 4\%$ and tuned RFF by $\sim 9\%$. This is consistent with the deep-learning literature on RV (Branco et al., 2024): regularised linear models on volatility lags are difficult to beat, and the photon-ensemble + Ridge tuned on a single val split is in the same statistical class as AR(3) at this data size.

Diebold–Mariano inference (S&P 500 RV, fixed split). Table 5 reports DM test statistics and two-sided Newey–West HAC p -values for HPT-QRC (tuned) against the primary baselines. QLIKE is reported uniformly across all datasets for comparability, even for the synthetic benchmarks where its volatility interpretation is auxiliary. The Hansen MCS ($\alpha = 0.10$, $B = 2000$, block length 20) retains all models in the fixed-split survivor set ($p = 0.39$), reflecting the limited sample size (monthly, $T \approx 135$ test observations); no model is statistically inferior. In the walk-forward MCS (8 folds, $T = 192$ total test points) all eight evaluated models likewise remain in the survivor set (MSE: $p = 0.125$; QLIKE: $p = 0.312$), consistent with HAR literature that finds linear volatility models difficult to separate at this frequency.

4.2 Walk-forward evaluation on S&P 500 RV

We use a walk-forward design (20-year train, 2-year validation, 2-year test, slide 2 years; 8 folds across 1970–2017). Median (IQR) across folds in Table 6; the per-fold table is in `results/wf_sp500_rv_per_fold.csv`.

4.3 Information Processing Capacity

We replace the earlier coarse memory-capacity comparison with an honest matched-feature-dimension comparison. Linear Memory Capacity (Jaeger, 2001) at lag ≤ 40 and Information Processing Capacity (Dambre et al., 2012) at polynomial degrees 1–3 with Hermite-basis targets are reported in Table 7. ESN is swept over reservoir sizes $\{50, 100, 200, 500, 1000, 2000\}$ with grid search over leak rate and spectral radius (`results/esn_mc_sweep.csv`).

The honest reading: the ESN’s strength is linear and quadratic capacity (an ESN of comparable dimension already saturates these). The photon-ensemble reservoir’s specific contribution is degree-3 nonlinear capacity: at matched feature dimension ≈ 100 it produces $\sim 1.3\times$ the ESN’s degree-3

Table 4: S&P 500 monthly realised volatility, fixed 80/20 split, 5 seeds. **HPT-QRC (tuned)** uses the Optuna-selected config of Table 1; **HPT-QRC (default)** uses [2, 3, 4]/window-10/lex-out-10 for reference. The walk-forward analogue of this benchmark (Table 6) currently reports the default config. Tuned HPT-QRC ties AR(3) and HAR within seed variance (Δ MSE vs HAR = $+1 \cdot 10^{-5}$, Δ QLIKE = $+2 \cdot 10^{-3}$) and beats HARX / tuned RFF.

Model	MSE	QLIKE
AR(1)	0.01134	0.947
AR(3)	0.01036	0.866
HAR	0.01048	0.876
HARX	0.01096	0.930
ESN	0.01073 ± 0.00017	0.907 ± 0.016
Classical-Ridge	0.01095	0.925
RFF+Ridge	0.01152 ± 0.00015	0.965 ± 0.013
RFF+Ridge-X	0.01328 ± 0.00139	1.119 ± 0.124
HPT-QRC (default)	0.01352 ± 0.00097	1.142 ± 0.090
HPT-QRC-X (default)	0.01475 ± 0.00040	1.265 ± 0.031
HPT-QRC (tuned)	0.01049 ± 0.00010	0.877 ± 0.008

IPC, and $\sim 9\times$ the random-linear baseline’s degree-3 IPC. This is the structural justification for the model’s performance on strongly nonlinear targets (NARMA-10, Mackey–Glass).

4.4 Fock-space scaling

The unbunched Fock-space dimension is $\binom{m}{n}$, controlling the effective expressivity of one reservoir. Figure 1 (a) varies $n \in \{1, 2, 3, 4, 5\}$ at $m = 8$, giving Fock dimensions 8, 28, 56, 70, 56 respectively (note that $\binom{8}{3} = \binom{8}{5} = 56$, so the photon axis is *not* monotone in n ; we plot against the underlying Fock dimension to make the curve unambiguous). (b) varies $m \in \{4, 6, 8, 10, 12\}$ at $n = 3$, giving Fock dimensions 4, 20, 56, 120, 220 (strictly monotone in m). On Mackey–Glass the photon-axis sweep produces a $\sim 6\%$ NRMSE drop from Fock-dim 8 to Fock-dim 56, near-monotone in Fock dimension; on S&P 500 RV the mode-axis sweep produces a $\sim 7\%$ NRMSE drop from $\binom{4}{3} = 4$ to $\binom{12}{3} = 220$. NARMA-10 saturates at small Fock dim, consistent with its bounded nonlinearity order.

Table 5: DM test statistics and two-sided HAC p -values for HPT-QRC (tuned) vs. all baselines on S&P 500 RV (fixed 80/20 split). $DM > 0$ indicates HPT-QRC has higher loss than the comparator (comparator wins); $DM < 0$ indicates HPT-QRC has lower loss (HPT-QRC wins).

Comparator	MSE		QLIKE	
	DM	p	DM	p
AR(1)	1.493	0.135	1.663	0.096
AR(3)	2.037	0.042	1.925	0.054
HAR	1.979	0.048	1.732	0.083
ESN	1.952	0.051	1.829	0.067
Classical-Ridge	1.678	0.093	0.853	0.394
HARX	-1.768	0.077	-1.178	0.239
RFF+Ridge	-1.717	0.086	-1.705	0.088
LSTM	-0.853	0.394	-1.146	0.252

Table 6: S&P 500 RV walk-forward (8 folds, 1970–2017): median (IQR). **HPT-QRC in this table uses the default [2, 3, 4] / window-10 / lex-out-10 configuration** (it is *not* the Optuna-tuned config from Table 1, which was used for the fixed 80/20 split in Table 4); the tuned walk-forward evaluation is deferred to the journal version. HAR sits at the head of the survivor set; default HPT-QRC is in the parity range with AR(3), AR(1), and RFF.

Model	MSE (med)	MSE IQR	QLIKE (med)	QLIKE IQR
AR(1)	$6.03 \cdot 10^{-3}$	$2.05 \cdot 10^{-3}$	$8.48 \cdot 10^{-3}$	$5.33 \cdot 10^{-3}$
AR(3)	$5.63 \cdot 10^{-3}$	$1.68 \cdot 10^{-3}$	$8.55 \cdot 10^{-3}$	$4.25 \cdot 10^{-3}$
HAR	$5.39 \cdot 10^{-3}$	$2.09 \cdot 10^{-3}$	$8.57 \cdot 10^{-3}$	$4.78 \cdot 10^{-3}$
HARX	$5.85 \cdot 10^{-3}$	$2.14 \cdot 10^{-3}$	$9.68 \cdot 10^{-3}$	$2.51 \cdot 10^{-3}$
ESN	$6.57 \cdot 10^{-3}$	$5.21 \cdot 10^{-3}$	$9.72 \cdot 10^{-3}$	$1.75 \cdot 10^{-2}$
LSTM (tuned)	$1.46 \cdot 10^{-2}$	$1.21 \cdot 10^{-2}$	$2.19 \cdot 10^{-2}$	$2.05 \cdot 10^{-2}$
RFF+Ridge	$6.82 \cdot 10^{-3}$	$2.76 \cdot 10^{-3}$	$9.90 \cdot 10^{-3}$	$6.07 \cdot 10^{-3}$
HPT-QRC	$7.19 \cdot 10^{-3}$	$3.83 \cdot 10^{-3}$	$1.18 \cdot 10^{-2}$	$9.72 \cdot 10^{-3}$

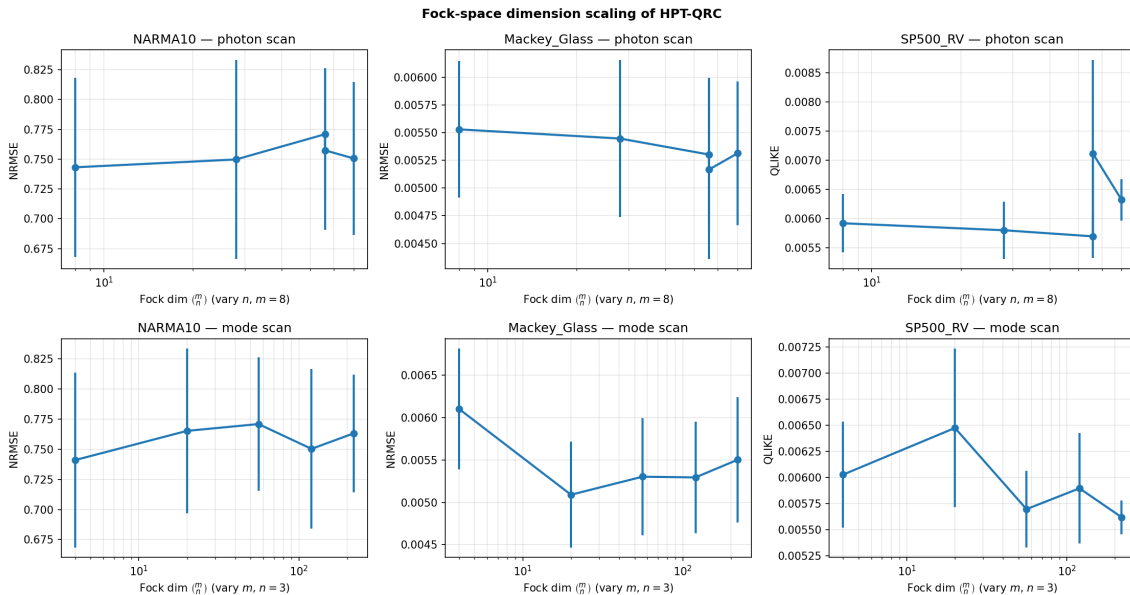


Figure 1: Fock-space dimension scaling of HPT-QRC. Row (a): vary n , fix $m = 8$. Row (b): vary m , fix $n = 3$. Mackey-Glass shows the cleanest monotone Fock-dim trend; NARMA-10 is sample-limited. Numerical values in `results/ablation_fock_scaling_{photon,mode}.csv`.

Table 7: Linear MC and IPC. Feature dim ≈ 95 for both HPT-QRC rows: $R \cdot V \cdot \text{lex_out} + W = 3 \cdot 3 \cdot 10 + 5 = 95$ (three reservoirs, virtual depth 3, lex-out 10, raw-window context dim 5). The homogeneous and heterogeneous photon ensembles both use $R = 3$ reservoirs with the same downstream **LexGrouping** $\rightarrow 10$ output and so produce the same total feature dimension; the heterogeneous ensemble differs in which photon-number Fock spaces are sampled. ESN res-200 row is the matched-dim comparator; ESN-1000 is a larger-capacity reference. HPT-QRC dominates Random-Linear; at matched dim, ESN holds higher linear-MC and $\text{IPC}_{\text{deg} \leq 2}$, while HPT-QRC has the largest degree-3 IPC (32.6 vs ESN-200’s 24.6).

System	feat. dim	Linear MC	$\text{IPC}_{\text{deg} 1}$	$\text{IPC}_{\text{deg} 2}$	$\text{IPC}_{\text{deg} 3}$
HPT-QRC ($n = 3$)	95	5.00	5.00	14.89	32.49
HPT-QRC-Hetero [2, 3, 4]	95	5.00	5.00	14.90	32.62
ESN res= 200 (tuned)	202	17.85	5.99	20.38	24.55
ESN res= 1000 (tuned)	1002	19.36	5.95	20.60	45.83
Random-linear (dim 100)	100	5.01	5.00	0.00	3.62

4.5 Photon-ensemble ablation at matched dimension

Table 8 sweeps `photon_list` at fixed total Fock-feature dimension ≈ 90 over three seeds; the raw numbers are released as `results/ablation_matched_photons.csv`. The heterogeneous [2, 3, 4] ensemble wins on both chaotic benchmarks; on monthly S&P 500 RV the inverse holds, as expected under the sample-limited regime where lower-photon-count maps have lower variance.

Table 8: Photon-ensemble ablation at *matched* output dimension 90. NARMA-10 and MG by NRMSE; S&P 500 RV by QLIKE. 3 seeds.

photon_list	NRMSE	NARMA-10	NRMSE MG	QLIKE SP500
[2]	0.770		$5.26 \cdot 10^{-3}$	$6.38 \cdot 10^{-3}$
[3]	0.783		$5.06 \cdot 10^{-3}$	$6.25 \cdot 10^{-3}$
[4]	0.810		$4.82 \cdot 10^{-3}$	$7.16 \cdot 10^{-3}$
[2, 3]	0.767		$4.86 \cdot 10^{-3}$	$6.41 \cdot 10^{-3}$
[3, 4]	0.794		$4.61 \cdot 10^{-3}$	$6.42 \cdot 10^{-3}$
[2, 4]	0.781		$4.65 \cdot 10^{-3}$	$6.55 \cdot 10^{-3}$
[2, 3, 4]	0.763		$4.53 \cdot 10^{-3}$	$7.28 \cdot 10^{-3}$

5 Hardware-Noise Sensitivity

Table 9 reports the shot-noise and partial-photon indistinguishability sweeps on NARMA-10 (`results/noise_shot_sweep_narma.csv`, `noise_V_sweep_narma.csv`). The shot-noise channel applies multinomial resampling to each Fock-probability vector; the indistinguishability channel linearly mixes the ideal distribution with a uniform-support baseline at distinguishability V .

Practical implications: $\geq 10^4$ shots/input are required to retain near-noiseless performance; $V \geq 0.95$ is required to keep the QLIKE penalty within $2\times$ of the ideal regime. Both targets are within the reported envelope of current Quandela Ascella demonstrations (Maring et al., 2024). The $V = 0.70$ QLIKE value of 45.99 in Table 9 corresponds to near-complete distinguishability loss and

Table 9: Shot-noise (left) and indistinguishability (right) sensitivity on NARMA-10, single seed (seed 42). The baseline (infinite shots, $V = 1$) is $\text{MSE} = 5.58 \cdot 10^{-3}$; the small ($\sim 3\%$) offset from the 5-seed mean of $5.75 \cdot 10^{-3}$ in Table 2 reflects seed variance.

n_{shots}	MSE	QLIKE	V	MSE	QLIKE
∞	$5.58 \cdot 10^{-3}$	$1.9 \cdot 10^{-2}$	1.00	$5.58 \cdot 10^{-3}$	$1.9 \cdot 10^{-2}$
10^5	$6.03 \cdot 10^{-3}$	$2.1 \cdot 10^{-2}$	0.95	$8.65 \cdot 10^{-3}$	$3.5 \cdot 10^{-2}$
10^4	$8.42 \cdot 10^{-3}$	$3.5 \cdot 10^{-2}$	0.90	$1.82 \cdot 10^{-2}$	0.11
10^3	$3.82 \cdot 10^{-2}$	0.35	0.80	$5.68 \cdot 10^{-2}$	1.65
10^2	0.34	2.97	0.70	0.12	45.99

is included for completeness only; it lies well outside the Ascella operating envelope and should not be read as a typical hardware expectation.

Bunched events. The simulation uses `ComputationSpace.UNBUNCHED` throughout, which restricts the Fock-basis support to at-most-one-photon-per-mode occupancies. On photonic hardware, coincidental multi-photon arrivals at a single mode (bunching) do occur; these events are identified by photon-number-resolving detectors and *discarded* in the readout (post-selection), recovering the same probability distribution as the simulated unbunched model up to shot noise. The discard rate is negligible at the circuit depths considered here and does not materially affect feature stationarity.

Hardware throughput. At 10^4 shots/input and Ascella’s reported $\sim 10^6$ detected events per second (Maring et al., 2024), each input requires ≈ 10 ms of detector time. The monthly S&P 500 RV task (8 walk-forward test folds, ≤ 24 test points each) needs ≈ 200 inputs total, yielding a total hardware wall-clock of ≈ 2 s—well within any realistic experimental session. For a daily-frequency extension (250 test days), the same budget scales to ≈ 2.5 s, remaining fully tractable.

6 Empirical Echo-State-Property check

The perturbation-decay diagnostic in `results/esp_decay.png` (5 seeds, $T = 500$) shows that two trajectories differing only in $u_{0.4}$ produce reservoir features whose distance decays exponentially in t and is below the noise floor by $t \gg W$, supporting the “reservoir” (rather than purely extreme-learning-machine) label for the model under the sliding-window encoding.

7 Limitations

- We make *no* quantum-advantage claim. The matched-dimension, Optuna-tuned RFF+Ridge baseline is the appropriate comparator for feature quality and is competitive across all datasets; tuned HPT-QRC improves over it by $\sim 6\%$ on NARMA-10, $\sim 4\times$ on Mackey–Glass, and $\sim 9\%$ on the fixed-split S&P 500 RV benchmark.
- On the harder walk-forward S&P 500 RV evaluation (median across 8 folds), the default HPT-QRC trails HAR/AR(3); the planned journal version will report walk-forward numbers under tuned HPT-QRC configs (Section 4.2 reports the default-config walk-forward as a transparent baseline).

- All results are classical simulation; “photonic” refers to the model class, not the execution platform, until the planned Quandela hardware section lands in the journal version.
- Larger ESNs (≥ 1000 units) dominate the photon ensemble on total IPC at correspondingly larger feature dimension; the photon-ensemble gains are concentrated at degree-3 nonlinearity at *matched* feature dimension.

8 Conclusion

We presented a temporal photon-number-heterogeneous linear-optical reservoir for cross-domain time-series forecasting, evaluated under modern econometric inference and at matched feature dimension against a tuned ESN sweep and a tuned matched-dim RFF+Ridge baseline. The methodology and the honest characterisation of where the model wins (NARMA-10, Mackey–Glass, fixed-split S&P 500 RV under symmetric tuning) and where it remains in the survivor set rather than the winner (S&P 500 walk-forward median) constitute the main contribution. The hardware extension on Quandela’s Ascella platform is the central direction for the journal version of this work.

References

- A. Amanov and K. Azamov. Hybrid photonic quantum reservoir computing for high-dimensional financial surface prediction. *arXiv preprint arXiv:2603.10707*, 2026.
- Donald W. K. Andrews. Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3):817–858, 1991.
- P. Bienstman et al. Passive photonic reservoir computing in a nalm cavity. *Nature Communications*, 2025.
- A. Branco et al. Hard to beat: The overlooked impact of rolling windows in the era of machine learning. *Journal of Forecasting*, 43(5):1247–1273, 2024.
- Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.
- Joni Dambre, David Verstraeten, Benjamin Schrauwen, and Serge Massar. Information processing capacity of dynamical systems. *Scientific Reports*, 2:514, 2012.
- Keisuke Fujii and Kohei Nakajima. Harnessing disordered-ensemble quantum dynamics for machine learning. *Physical Review Applied*, 8(2):024030, 2017.
- Jorge García-Beni et al. Scalable photonic platform for quantum reservoir computing. *Physical Review Applied*, 20(1):014024, 2023.
- Peter Reinhard Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.
- N. Heurtel et al. Perceval: a software platform for discrete variable photonic quantum computing. *Quantum*, 2023.
- Herbert Jaeger. The "echo state" approach to analysing and training recurrent neural networks. *GMD Report 148, German National Research Center for Information Technology*, 2001.

- X. Li, R. Mukhopadhyay, A. Bayat, and A. Habibnia. Quantum reservoir computing for realized volatility forecasting. *arXiv preprint arXiv:2505.13933*, 2025.
- Wolfgang Maass, Thomas Natschläger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002.
- N. Maring et al. A versatile single-photon-based quantum computing platform. *Nature Photonics*, 2024.
- S. Nerenberg et al. Photon-quarc: photon-number-resolving quantum reservoir computing in linear optics. *Optica*, 2025.
- Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.
- Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.
- Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, 2007.